

Project Plan Documentation

Project Title

FarmVerse Precision Agriculture Management

1. Introduction

FarmVerse Nexus is a Full Stack Java web application that helps farmers manage farming activities digitally. It enables users to manage farms, crops, irrigation, expenses, harvests, weather updates, and reports through a simple and secure platform.

2. Purpose

- Digitize farm management.
- Reduce manual record keeping.
- Track farming activities efficiently.
- Monitor expenses and income.
- Improve productivity through better decision-making.

3. Objectives

- Develop a secure Full Stack Java application.
- Manage farm and crop details.
- Track irrigation and fertilizer usage.
- Record expenses and harvest information.
- Generate reports and analytics.
- Provide weather updates and notifications.

4. Scope

Farmer

- Register & Login
- Manage Farms
- Manage Crops
- Schedule Irrigation
- Track Expenses
- Record Harvest

- View Weather
- Generate Reports

Admin

- Manage Users
- Monitor Farm Data
- View Reports
- Manage Notifications
- Dashboard Analytics

5. Technologies Used

Frontend	Backend	Database Tools
HTML, CSS, Bootstrap, JavaScript	Java, Spring Boot, Spring MVC, REST API	MySQL Hibernate, Spring Data JPA, Maven, GitHub, Postman

6. Project Modules

User Management

- Registration
- Login
- Profile Management
- Password Reset

Farm Management

- Add/Edit/Delete Farms
- Soil & Land Details
- Water Source

Crop Management

- Crop Details
- Growth Tracking
- Harvest Schedule

Irrigation Management

- Schedule Irrigation
- Water Usage
- Irrigation History

Fertilizer Management

- Fertilizer & Pesticide Records
- Usage History
- Cost Tracking

Expense Management

- Farming Expenses
- Income Tracking
- Profit/Loss Calculation

Harvest Management

- Harvest Details
- Selling Price
- Income Report

Weather & Notification

- Weather Updates
- Irrigation Alerts
- Farming Reminders

Reports & Analytics

- Crop Reports
- Expense Reports
- Harvest Reports
- Dashboard Charts

7. Database Tables

- Users
- Farms
- Crops
- Irrigation
- Fertilizers
- Expenses
- Harvest
- Weather
- Notifications

- Reports

8. Functional Requirements

- Secure Login
- Farm Management
- Crop Tracking
- Expense Management
- Harvest Recording
- Weather Updates
- Report Generation

9. Non-Functional Requirements

- Secure
- Fast
- Responsive
- User-Friendly
- Reliable
- Scalable

10. Future Enhancements

- AI Crop Disease Detection
- IoT Sensor Integration
- Mobile Application
- SMS Alerts
- Drone Monitoring
- Yield Prediction